

ACTU 301
Life Contingencies 1

Life Insurance Benefits

Lecture 4

Introduction to life insurance functions

A life assurance (or life insurance) policy is a contract which promises to pay a specified sum, b say, on the death of a given life (the life assured) at any future time, or between certain specified dates.

The death benefit b is called the sum assured/insured; it may be level (constant) or it may increase or decrease in a manner specified in the contract.

Policies under which payments depend on the death or survival of more than one life may also be issued.

- ❖ We shall for now consider contracts issued on the life of one person.
- ❖ We shall further focus on level benefit policies which pay the same amount of benefit regardless of the time at which the benefit is paid.
- ❖ Benefit payments can be either made at the time the contingent event occurs or at some set time like the end of the year in which the contingent event occurs.
- ❖ We shall value benefits payments contingent on death or survival of a policyholder by combining time value of money and survival models.
- ❖ In particular, we shall use the future lifetime random variable T_x to determine the value of a benefit payable immediately on the contingent event of an individual, and the curtate future lifetime random variable K_x for when a payment is made at the end of the year in which the contingent event occurs.
- ❖ We shall develop formulas for the expected present value of the benefits paid by an insurance company for a number of different contract types.

Financial mathematics review

The valuation of benefits involve some interest theory functions.

Given an annual effective interest rate i , the annual interest rate convertible m thly satisfies:

$$1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m$$

and if the interest is compounded continuously we use the force of interest which is given by

$$\delta = \log(1 + i)$$

The annual effective discount rate can be defined in terms of the discount factor v as

$$d = iv = 1 - v \quad \text{where} \quad v = \frac{1}{1 + i} = e^{-\delta}$$

Finally, given d , the annual discount rate convertible m thly satisfies

$$1 - d = \left(1 - \frac{d^{(m)}}{m}\right)^m$$

I. Whole Life Insurance

Whole Life Insurance payable at the moment of death

By a whole life insurance we mean an insurance that makes a benefit payment at the time of death of the insured person, no matter when that time might be.

The models will be developed with a benefit function b and a discount function v ;

To simplify calculations, we assume for now that the benefit amount is equal to GH¢1 (called unit insurance).

So that the benefit function is

$$b_{T_x} = 1, \text{ for all } T_x \geq 0.$$

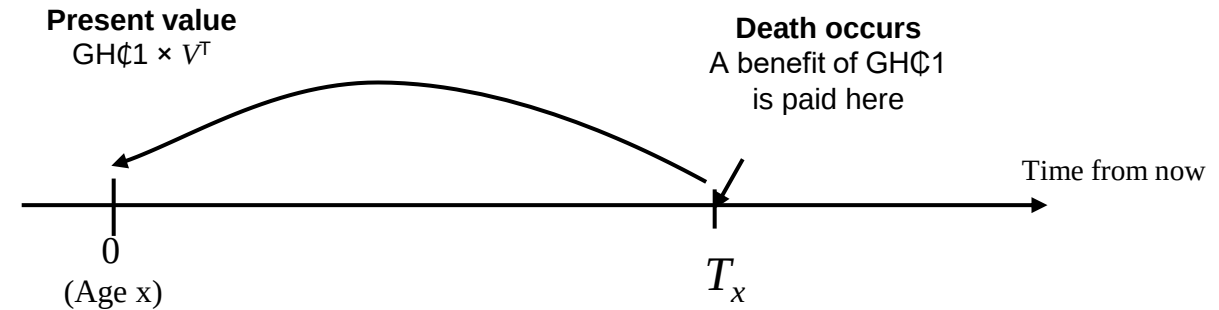
Multiplying the benefit function with the discount factor for a period of T_x years, we obtain the present value random variable

$$Z = v^{T_x} \quad \text{for all } T_x \geq 0.$$

Z denotes the present value random variable for a unit-benefit insurance policy.

Whole Life Insurance payable at the moment of death

The diagram below illustrates the mechanism of a unit-benefit continuous whole life insurance:



Whole Life Insurance payable at the moment of death

The average cost of a whole life insurance is referred to as the actuarial present value (APV) of the random variable Z .

The average cost of a whole life insurance policy is represented as the expected value of Z (i.e. expected present value)

Thus The average cost = APV = $E(Z)$

For a life aged x , this average cost $E(Z)$ would be denoted by $\bar{A}_x$

$$\bar{A}_x = E(Z)$$

In the symbol,

- The A stands for the APV of a life insurance,
- The x indicates the age of the policyholder at the beginning of the contract.
- The bar over the A indicates that this average cost is calculated on a continuous basis (i.e. at the time of death)

We also say that

- ❖ The quantity $\bar{A}_x$ is also called the *net single premium* of a whole life insurance with a unit payment made at the moment of the death.
- ❖ By net we mean that the premium is not loaded with any expenses, and;
- ❖ By single, we mean that the premium is paid as a single lump sum at the inception of the contract.

Whole Life Insurance payable at the moment of death

Once we know how to deal with insurances paying a unit, we can deal with insurances making a general payment.

The actuarial present value of a whole life insurance with payment b is $b\bar{A}_x$.

To calculate the expected value, we simply integrate out the product of Z and the density function of the underlying random variable T_x , that is,

$$E(Z) = \bar{A}_x = E(v^t)$$

$$E(Z) = \bar{A}_x = \int_0^{\infty} v^t f_x(t) dt$$

$$\bar{A}_x = \int_0^{\infty} v^t {}_t p_x \mu_{x+t} dt$$

$$\bar{A}_x = \int_0^{\infty} e^{-\delta t} {}_t p_x \mu_{x+t} dt$$

if there is a limiting age, we replace ∞ with $\omega - x$

To calculate the variance of Z , we can make use of the identity

$$\text{Var}(Z) = E(Z^2) - [E(Z)]^2.$$

where

$$E(Z^2) = \int_0^{\infty} (v^t)^2 {}_t p_x \mu_{x+t} dt$$
$$= \int_0^{\infty} e^{-2\delta t} {}_t p_x \mu_{x+t} dt.$$

Whole Life Insurance payable at the moment of death

- ❖ We observe that the formula for $E(Z^2)$ is the same as that for $E(Z)$, except that δ is replaced with 2δ .
- ❖ To obtain $E(Z^2)$, we simply evaluate $E(Z)$ at two times the original force of interest.

- ❖ We denote $E(Z^2)$ for a continuous unit-benefit whole life insurance by ${}^2\bar{A}_x$.

$$E(Z^2) = {}^2\bar{A}_x$$

- ❖ The superscript 2, indicates that the quantity is $E(Z)$ calculated at 2δ .

$$Var(Z) = {}^2\bar{A}_x - (\bar{A}_x)^2$$

The result that $E(Z^2)$ is $E(Z)$ evaluated at 2δ is applicable to not only a unit-benefit whole life insurance, but also any insurance policy with $b_{Tx} = 0$ or 1 for all T_x .

Whole Life Insurance payable at the end of the year of death

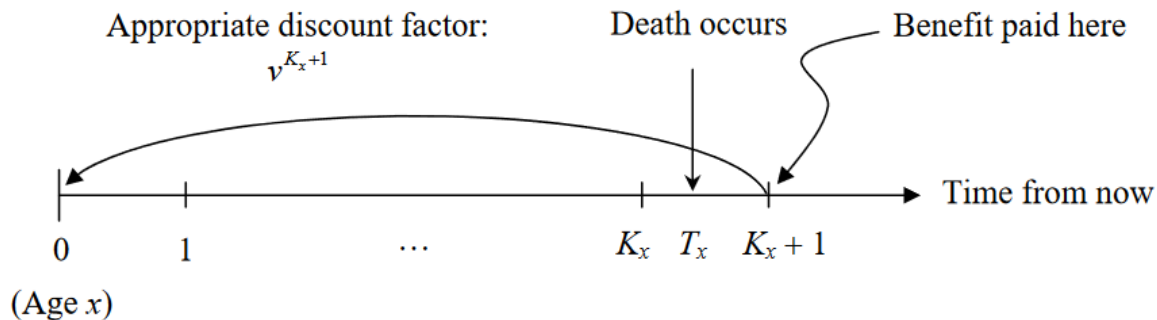
Suppose that the death benefit is payable at the end of the year of death (years being measured from the date of issue of the policy).

The present value random variable for this benefit is

$$Z = v^{K_x+1} \quad \text{for } K_x = 0, 1, 2, \dots$$

Here, Z depends on a discrete random variable (K_x).

The timing of the death benefit is illustrated by the following diagram.



Whole Life Insurance payable at the end of the year of death

This means that $E(Z)$ should be calculated as a summation instead of an integral, and it should be based on the probability function of K_x

The Actuarial Present Value of this policy for life (x) with one unit in benefits payable at the end of the year of death is given by the formula

$$E(Z) = \sum_{k=0}^{\infty} v^{k+1} P(K_x = k)$$

$$E(Z) = \sum_{k=0}^{\infty} v^{k+1} {}_k|q_x$$

$$E(Z) = \sum_{k=0}^{\infty} v^{k+1} {}_k p_x q_{x+k}$$

We denote $E(Z)$ by A_x

$$A_x = \sum_{k=0}^{\infty} v^{k+1} {}_k p_x q_{x+k}$$

Whole Life Insurance payable at the end of the year of death

The second moment of Z is denoted by 2A_x and is given by

$${}^2A_x = \sum_{k=0}^{\infty} v^{2(k+1)} {}_k p_x q_{x+k}$$

The variance of Z is then

$$\text{Var}(Z) = {}^2A_x - (A_x)^2$$

Whole Life Insurance payable at the end of the year of death

Example

Let the remaining lifetime at birth random variable X be uniform on $[0,100]$.

Let Z_{30} be the contingent payment random variable for a life aged $x = 30$.

Find A_{30} , ${}^2A_{30}$, and $\text{Var}(Z_{30})$ if $\delta = 0.05$.

$$s(x) = 1 - \frac{x}{100}.$$

$${}_k p_{30} = \frac{s(30+k)}{s(30)} = \frac{1 - \frac{30+k}{100}}{1 - \frac{30}{100}} = 1 - \frac{k}{70}$$

$$q_{30+k} = 1 - \frac{s(31+k)}{s(30+k)} = \frac{1}{70-k}$$

$$A_{30} = \sum_{k=0}^{\infty} \frac{e^{-0.05(k+1)}}{70} = \frac{e^{-0.05}}{70(1 - e^{-0.05})} = 0.2786$$

$${}^2A_{30} = \sum_{k=0}^{\infty} \frac{e^{-0.10(k+1)}}{70} = \frac{e^{-0.10}}{70(1 - e^{-0.10})} = 0.1358$$

$$\text{Var}(Z_{30}) = 0.1358 - 0.2786^2 = 0.0582$$

II. Term (or temporary) assurance/Term life insurance

An n -year term life insurance is a policy where the payment is made only when the insured person dies within n years of the policy's issue. That is, if the insured survives n years, no benefit is paid.

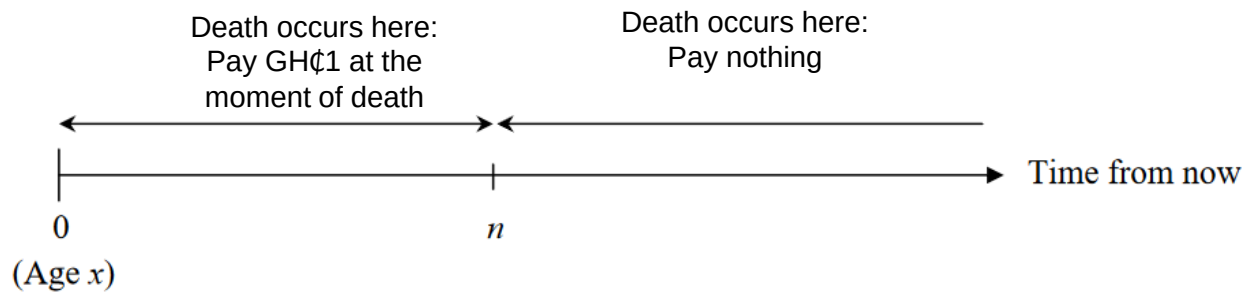
That is, a term (or temporary) assurance contract pays a sum assured on the death of (x) within the term of the policy, which we usually write as n years.

Consider an n -year term insurance with unit benefit payable immediately on the death of an individual. If the individual is currently aged x , the benefit will be paid in T_x years given $T_x \leq n$, therefore, to get the present value of a unit benefit the payment must be discounted by T_x years.

The benefit function is given by

$$b_{T_x} = \begin{cases} 1, & T_x \leq n \\ 0, & T_x > n \end{cases}$$

The payout from a unit-benefit n -year term life insurance is illustrated in the following diagram.



n -year term insurance payable immediately on the death

If the benefit is payable immediately on death, the present value (as a random variable) is

$$Z = \begin{cases} v^{T_x} = e^{-\delta T_x} & \text{if } T_x \leq n \\ 0 & \text{if } T_x > n \end{cases}$$

The expected value of Z , sometimes referred to as the actuarial present value (APV) and denoted by $\bar{A}_{x:n}^1$ is

$$\bar{A}_{x:n}^1 = E(Z) = \int_0^n e^{-\delta t} f_x(t) dt = \int_0^n e^{-\delta t} {}_t p_x \mu_{x+t} dt$$

$$\bar{A}_{x:n}^1 = E(Z) = \int_0^n v^t f_x(t) dt = \int_0^n v^t {}_t p_x \mu_{x+t} dt$$

In actuarial notation,

- The bar above the A denotes that the benefit is payable immediately on death,
- The '1' above the x indicates that for the benefit to be paid the life must **die** before the end of the n -year term.
- The subscript $\overline{n}$ indicates that coverage is provided for at most n years.

***n*-year term insurance payable immediately on the death**

This policy satisfies the criterion that

$$b_{T_x} = 0 \text{ or } 1 \text{ for all } T_x$$

$E(Z^2)$ is $E(Z)$ evaluated at 2δ is applicable

$$\text{Hence, we have } E(Z^2) = {}^2\bar{A}_{x:\overline{n}|}^1$$

That is $\bar{A}_{x:\overline{n}|}^1$ evaluated at 2δ

$$\text{Var}(Z) = {}^2\bar{A}_{x:\overline{n}|}^1 - (\bar{A}_{x:\overline{n}|}^1)^2$$

Example

The age-at-death random variable is exponential with parameter 0.05. A life aged 35 buys a 25-year life insurance policy that pays 1 upon death. Assume an annual effective interest rate of 7%, find the actuarial present value of this policy.

Solution

The actuarial present value of this policy is

$$E(Z) = \bar{A}_{35:\overline{25}|}^1 = \int_0^{25} e^{-\delta t} f_{35}(t) dt = \int_0^{25} v^t f_{35}(t) dt$$

Given $X=35$, $n=25$ $i=0.07$

$$\bar{A}_{35:\overline{25}|}^1 = \int_0^{25} v^t f_{35}(t) dt$$

Pdf of exponential distribution $f(t) = \lambda e^{-\lambda t}$, $t > 0$

$$f_{35}(t) = f_{35}(t) = \lambda e^{-\lambda t} = 0.05 e^{-0.05t}$$

$$v^t = (1+i)^{-t}$$

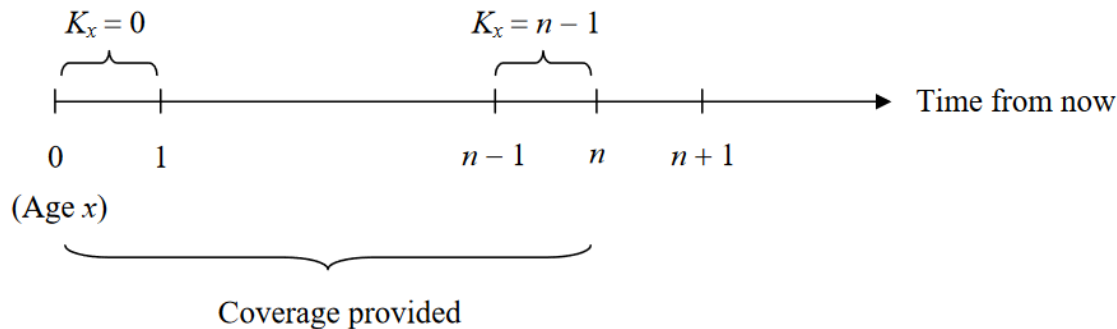
On substitution we obtain

$$E(Z) = \bar{A}_{35:\overline{25}|}^1 = \int_0^{25} (1.07)^{-t} e^{-0.05t} (0.05) dt = 0.4025$$

Term Life Insurance payable at the end of the year of death

If the sum assured is payable at the end of year of death (if this occurs within the n year term), we have the present value

$$Z = \begin{cases} v^{K+1} & \text{if } K < n \\ 0 & \text{if } K \geq n \end{cases}$$



From the diagram above, you can see that the summation should run from $k = 0$ to $k = n - 1$

Hence, $E(Z)$ for a unit-benefit term life insurance is given by

$$E(Z) = \sum_{k=0}^{n-1} v^{k+1} {}_k p_x q_{x+k}$$

Term Life Insurance payable at the end of the year of death

$$E(Z) = \sum_{k=0}^{n-1} v^{k+1} {}_k p_x q_{x+k}$$

We denote $E(Z)$ by $A_{x:\overline{n}|}^1$

$$A_{x:\overline{n}|}^1 = \sum_{k=0}^{n-1} v^{k+1} {}_k p_x q_{x+k}$$

We should expect $E(Z)$ for a term life insurance to be the same as that for a whole life insurance, except that the upper limit of the summation is no longer ∞ .

- ❖ As before $\overline{n}|$ indicates that this policy will last for at most n years,
- ❖ The 1 above x indicates that the policy is a term life insurance.
- ❖ There is no bar above A , because the policy is discrete rather than continuous.

$E(Z^2)$ is the same as $E(Z)$ evaluated at 2δ

Hence, we have $E(Z^2) = {}^2A_{x:\overline{n}|}^1$

$$\text{Var}(Z) = {}^2A_{x:\overline{n}|}^1 - (A_{x:\overline{n}|}^1)^2$$

Term Life Insurance payable at the end of the year of death

Example 1

Let the remaining lifetime at birth random variable X be uniform on $[0,100]$.

Let $Z_{30:\overline{10}|}^1$ be the contingent payment random variable for a life aged $x = 30$.

Find $A_{30:\overline{10}|}^1$, ${}^2A_{30:\overline{10}|}^1$, and $\text{Var}(Z_{30:\overline{10}|}^1)$ if $\delta = 0.05$.

For this problem, recall

$$s(x) = 1 - \frac{x}{100}$$

We have

$${}_k p_{30} = \frac{s(30+k)}{s(30)} = \frac{1 - \frac{30+k}{100}}{1 - \frac{30}{100}} = 1 - \frac{k}{70}$$

$$q_{30+k} = 1 - \frac{s(31+k)}{s(30+k)} = \frac{1}{70-k}$$

$$A_{30:\overline{10}|}^1 = \sum_{k=0}^9 \frac{e^{-0.05(k+1)}}{70} = \frac{e^{-0.05}(1 - e^{-0.50})}{70(1 - e^{-0.05})} = 0.1096$$

$${}^2A_{30:\overline{10}|}^1 = \sum_{k=0}^9 \frac{e^{-0.10(k+1)}}{70} = \frac{e^{-0.10}(1 - e^{-1})}{70(1 - e^{-0.10})} = 0.0859$$

$$\text{Var}(Z_{30:\overline{10}|}^1) = 0.0859 - 0.1096^2 = 0.0739$$

Example

An individual currently aged 40 purchases a 15-year term insurance that pays a benefit of GH¢50,000 immediately on death. If the force of interest is $\delta = 0.05$ and

$$S_0(x) = 1 - \frac{x}{100} \quad 0 \leq x \leq 100$$

- (i) Find the APV of the death benefit.
- (ii) Find the standard deviation of the present value random variable.

Solution

- (i) The APV to be calculated is

$$\bar{A}_{40:\overline{15}|}^1 = E(Z) = \int_0^{15} e^{-\delta t} f_{40}(t) dt = \int_0^{15} e^{-\delta t} {}_t p_{40} \mu_{40+t} dt$$

By applying

$$\bar{A}_{x:n}^1 = E(Z) = \int_0^n e^{-\delta t} f_x(t) dt = \int_0^n e^{-\delta t} {}_t p_x \mu_{x+t} dt$$

$${}_t p_x = \frac{S_0(x+t)}{S_0(x)} = \frac{100 - x - t}{100} \cdot \frac{100}{100 - x} = 1 - \frac{t}{100 - x}$$

Hence for a life aged 40, ${}_t p_{40} = 1 - \frac{t}{60}$

Since we have not been given the force of mortality, it is much easier to compute the APV of the benefit using the formula involving the density, where

$$f_{40}(t) = -\frac{d}{dt} {}_t p_{40} = \frac{1}{60}$$

Recall $f_x(t) = -\frac{d {}_t p_x}{dt}$

The APV of a unit benefit is therefore

$$\bar{A}_{40:\overline{15}|}^1 = E(Z) = \int_0^{15} e^{-\delta t} f_{40}(t) dt$$

$$\begin{aligned} \bar{A}_{40:\overline{15}|}^1 &= \int_0^{15} e^{-\delta t} f_{40}(t) dt = \int_0^{15} e^{-\delta t} \frac{1}{60} dt \\ &= -\frac{1}{60\delta} [e^{-\delta t}]_0^{15} \\ &= \frac{1}{60\delta} (1 - e^{-15\delta}) = 0.1759 \end{aligned}$$

However, since the policy pays GH¢50,000 on death, the APV of the benefit is

$$\text{GH¢ } 50,000 \bar{A}_{40:\overline{15}|}^1 = \text{GH¢ } 50,000 \times 0.1759 = \text{GH¢ } 8,793.89$$

Using the rule of moments, the second moment of the present value random variable is

$$E[Z^2] = \frac{1}{120\delta} (1 - e^{-30\delta}) = 0.1295$$

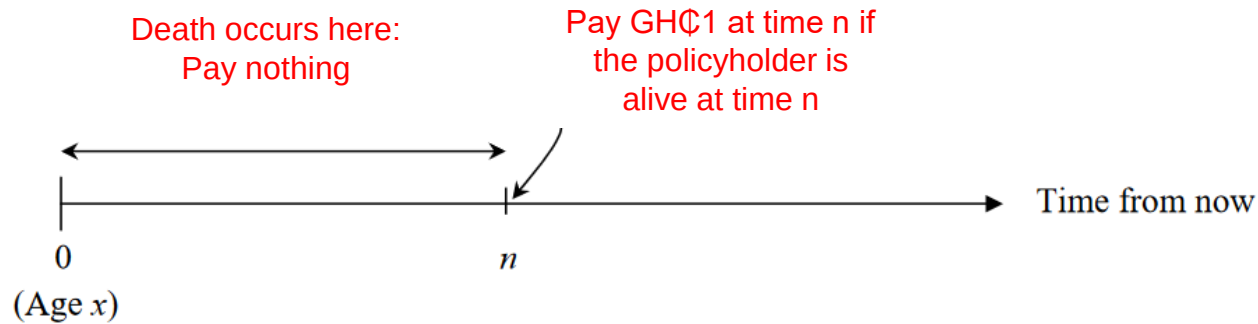
When finding the variance we need to make sure we find it for the total benefit and not just a unit benefit

$$\begin{aligned}\text{Var}[50000Z] &= 50000^2 \text{Var}[Z] = 50000^2 ({}^2\bar{A}_{40:\overline{15}|}^1 - (\bar{A}_{40:\overline{15}|}^1)^2) \\ &= 50000^2 (0.1295 - 0.1759^2) \\ &= 246,397,975\end{aligned}$$

III. Pure Endowment

An n -year pure endowment provides a payment at the end of n years if the policyholder survives, but makes no payment if the policyholder dies within n years.

If we assume the benefit amount is GH¢1. The payout from a unit benefit n -year pure endowment is illustrated in the diagram below



The benefit function is given by
$$b_{T_x} = \begin{cases} 0, & T_x \leq n \\ 1, & T_x > n \end{cases}$$

For an n -year pure endowment insurance a benefit is paid if the individual dies after n years, that is $T_x \geq n$. Since the benefit is only ever paid after n years, the present value random variable is

$$Z = \begin{cases} 0, & T_x \leq n \\ v^n, & T_x > n \end{cases}$$

Z is a discrete random variable that takes only two values, 0 with probability ${}_nq_x$, and v^n with probability ${}_np_x$, therefore, the expected value of Z is given by

$$E(Z) = 0 \times \Pr(T \leq n) + v^n \times \Pr(T > n) = v^n {}_np_x$$

We denote $E(Z)$ for a unit-benefit pure endowment by $A_{x:\overline{n}|}^1$

$$A_{x:\overline{n}|}^1 = E(Z) = v^n \Pr(T_x > n) = v^n {}_np_x$$

- The 1 above $\overline{n}|$ indicates that the policy is a pure endowment.(i.e. for the benefit to be paid the term must expire before the life).
- We do not write a bar above A , because the payment is made at time n instead of time T_x (as there only exists a discrete time version)

There are two actuarial notations for the actuarial present value:

- ❖ It may be denoted by
- ❖ This notation leads to a possibility of confusion between the term and pure endowment insurance policies. For this reason, the APV of the pure endowment insurance is often denoted by ${}_nE_x$.

$$A_{x:n}^1 = {}_nE_x = v^n {}_np_x$$

The second moment of Z can be calculated in the same way as for any discrete random variable

$$E(Z^2) = {}^2A_{x:\overline{n}|}^1 = v^{2n} {}_n p_x$$

$$\text{Var}(Z) = E(Z^2) - [E(Z)]^2$$

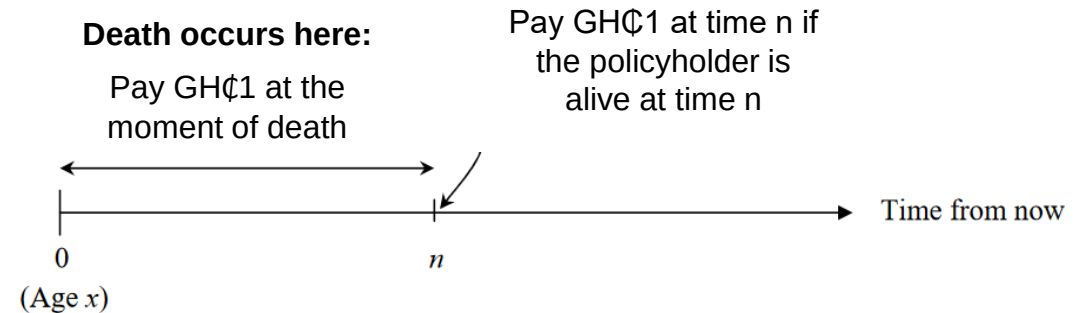
$$\begin{aligned}\text{Var}[Z] &= {}^2A_{x:\overline{n}|}^1 - (A_{x:\overline{n}|}^1)^2 \\ &= v^{2n} {}_n p_x - (v^n {}_n p_x)^2 \\ &= v^{2n} {}_n p_x {}_n q_x.\end{aligned}$$

IV. Endowment Insurance

An n -year endowment insurance provides an amount to be paid at the moment of death if death occurs in the next n years or at the end of year n if the policyholder survives to that time.

As before, we assume that the benefit amount is GH¢1.

The payout from a unit-benefit n -year endowment insurance is illustrated in the diagram below



The benefit function is $b_{T_x} = 1$ for all T_x .

Multiplying it with appropriate discount factors, we obtain the present value random variable

$$Z = \begin{cases} v^{T_x}, & T_x \leq n \\ v^n, & T_x > n \end{cases}$$

Endowment payable at the moment of death

It can be observed that an n-year endowment insurance is simply a combination of an n-year term insurance and an n-year pure endowment.

It follows that $E(Z)$ for an n-year endowment insurance can be expressed as

$$E(Z) = \bar{A}_{x:\overline{n}|}^1 + A_{x:\overline{n}|}^1$$

We denote $E(Z)$ by $\bar{A}_{x:\overline{n}|}$

$$\bar{A}_{x:n} = \bar{A}_{x:n}^1 + A_{x:n}^1$$

Note that there is no 1 above either x or $\overline{n}|$.

This policy satisfies the criterion that

$b_{T_x} = 0$ or 1 for all T_x $E(Z^2)$ is $E(Z)$ evaluated at 2δ is applicable

Hence, we have $E(Z^2) = {}^2\bar{A}_{x:\overline{n}|}$ That is $\bar{A}_{x:\overline{n}|}$ evaluated at 2δ

$${}^2\bar{A}_{x:n} = {}^2\bar{A}_{x:n}^1 + v^{2n} {}_n p_x$$

$$\text{Var}(Z) = {}^2\bar{A}_{x:\overline{n}|} - \bar{A}_{x:\overline{n}|}^2$$

Endowment payable at the end of the year of death

Similarly, if the death benefit is payable at the end of the year of death, the present value is

$$Z = \begin{cases} v^{K+1} & \text{if } K < n \\ v^n & \text{if } K \geq n \end{cases}$$

$$E(Z) = \sum_{k=0}^{n-1} v^{k+1} {}_k p_x q_{x+k} + v^n \Pr(K \geq n)$$

$$E(Z) = \sum_{k=0}^{n-1} v^{k+1} {}_k p_x q_{x+k} + v^n {}_n p_x$$

Recall that an n-year endowment insurance is just a combination of an n-year term life insurance and an n-year pure endowment. Hence, we have

$$E(Z) = \bar{A}_{x:\overline{n}|}^1 + A_{x:\overline{n}|}^1$$

We denote $E(Z)$ by $\bar{A}_{x:\overline{n}|}$ Hence we have

$$A_{x:n} = A_{x:n}^1 + A_{x:n}^1$$

$E(Z^2)$ is $E(Z)$ evaluated at 2δ is applicable

$$\text{Var}(Z) = {}^2\bar{A}_{x:\overline{n}|} - \bar{A}_{x:\overline{n}|}^2$$

V. Deferred Life Insurance

A deferred assurance provides a benefit on the death of (x) if this occurs after a certain period, called the deferred period (or, more correctly, the period of deferment), which we usually write as n years.

If the benefit is payable immediately on death of (x) after the deferred period has elapsed, the present value is

$$Z = \begin{cases} v^{T_x} = e^{-\delta T_x} & \text{if } T_x \geq n \\ 0 & \text{if } T_x < n \end{cases}$$

The APV of a deferred insurance benefit is given as:

$$E(Z) = {}_n|A_x = \int_n^{\infty} v^t f_x(t) dt = \int_n^{\infty} e^{-\delta t} f_x(t) dt$$

We denote $E(Z)$ by ${}_n|\bar{A}_x$

$${}_n|A_x = \int_n^{\infty} v^t {}_t p_x \mu_{x+t} dt = \int_n^{\infty} e^{-\delta t} {}_t p_x \mu_{x+t} dt$$

$${}_n|\bar{A}_x = v^n {}_n p_x \bar{A}_{x+n}$$

The variance of Z is $\text{Var}[Z] = {}^2_n|\bar{A}_x - ({}_n|\bar{A}_x)^2$

if the sum assured is payable at the end of year of death

$$Z = \begin{cases} v^{K+1} & \text{if } K \geq n \\ 0 & \text{if } K < n \end{cases}$$

The corresponding A.P.V.'s are written as

$$E(Z) = \sum_{k=n}^{\infty} v^{k+1} {}_k|q_{x+k} = \sum_{k=n}^{\infty} v^{k+1} {}_k p_x q_{x+k}$$

We denote $E(Z)$ by ${}_n|A_x$.

$$E(Z) = {}_n|A_x = \sum_{k=n}^{\infty} v^{k+1} {}_k|q_{x+k} = \sum_{k=n}^{\infty} v^{k+1} {}_k p_x q_{x+k}$$

$$E(Z) = {}_n|A_x = v^n {}_n p_x A_{x+n}$$

$$\text{Var}(Z) = {}^2_n|A_x - ({}_n|A_x)^2$$

The two relationships that we previously derived for continuous insurance functions also apply in the discrete case, that is

$$A_x = A_{x:\overline{n}|}^1 + {}_n|A_x$$

Recursive Relationships

We now state and prove the recursive formulae for the whole life, term, deferred, and endowment insurance functions.

$$\boxed{A_x = vq_x + vp_x A_{x+1}} \quad \text{whole life}$$

Proof:

$$\begin{aligned} A_x &= \sum_{k=0}^{\infty} v^{k+1} {}_k|q_x = \sum_{k=0}^{\infty} v^{k+1} {}_k p_x \cdot q_{x+k} \\ &= vq_x + \sum_{k=1}^{\infty} v^{k+1} {}_k p_x \cdot q_{x+k} \\ &= vq_x + vp_x \sum_{k=1}^{\infty} v^k {}_{k-1} p_{x+1} \cdot q_{x+k} \\ &= vq_x + vp_x \sum_{j=0}^{\infty} v^{j+1} {}_j p_{x+1} q_{x+j+1} \quad (\text{using the substitution } j = k - 1) \\ &= vq_x + vp_x \sum_{j=0}^{\infty} v^{j+1} {}_j|q_{x+1} = vq_x + vp_x A_{x+1} \end{aligned}$$

Recursive Relationships

$$\boxed{A_{x:\overline{n}|}^1 = vq_x + vp_x A_{x+1:\overline{n-1}|}^1} \quad \text{Term}$$

Proof:

$$\begin{aligned} A_{x:\overline{n}|}^1 &= \sum_{k=0}^{n-1} v^{k+1} {}_k|q_x = \sum_{k=0}^{n-1} v^{k+1} {}_k p_x \cdot q_{x+k} \\ &= vq_x + \sum_{k=1}^{n-1} v^{k+1} {}_k p_x \cdot q_{x+k} \\ &= vq_x + vp_x \sum_{k=1}^{n-1} v^k {}_{k-1} p_{x+1} \cdot q_{x+k} \\ &= vq_x + vp_x \sum_{j=0}^{n-2} v^{j+1} {}_j p_{x+1} \cdot q_{x+j+1} \quad (\text{using the substitution } j = k - 1) \\ &= vq_x + vp_x \sum_{j=0}^{n-2} v^{j+1} {}_j|q_{x+1} = vq_x + vp_x A_{x+1:\overline{n-1}|}^1 \end{aligned}$$

Recursive Relationships

$$\boxed{{}_n|A_x = v^{n+1}{}_np_x \cdot q_{x+n} + {}_{n+1}|A_x} \quad \text{Deferred}$$

Proof:

$$\begin{aligned} {}_n|A_x &= \sum_{k=n}^{\infty} v^{k+1}{}_k|q_x = \sum_{k=n}^{\infty} v^{k+1}{}_kp_x \cdot q_{x+k} \\ &= v^{n+1}{}_np_x \cdot q_{x+n} + \sum_{k=n+1}^{\infty} v^{k+1}{}_kp_x \cdot q_{x+k} \\ &= v^{n+1}{}_np_x \cdot q_{x+n} + {}_{n+1}|A_x \end{aligned}$$

Recursive Relationships

$$\boxed{A_{x:\overline{n}|} = vq_x + vp_x A_{x+1:\overline{n-1}|}} \quad \text{Endowment}$$

Proof:

$$\begin{aligned} A_{x:\overline{n}|} &= A_{x:\overline{n}|}^1 + {}_nE_x = \sum_{k=0}^{n-1} v^{k+1}{}_k|q_x + v^n{}_np_x = \sum_{k=0}^{n-1} v^{k+1}{}_kp_x \cdot q_{x+k} + v^n{}_np_x \\ &= vq_x + \sum_{k=1}^{n-1} v^{k+1}{}_kp_x \cdot q_{x+k} + v^n{}_np_x \\ &= vq_x + vp_x \left(\sum_{k=1}^{n-1} v^k{}_{k-1}p_{x+1} \cdot q_{x+k} + v^{n-1}{}_{n-1}p_{x+1} \right) \\ &= vq_x + vp_x \left(\sum_{j=0}^{n-2} v^{j+1}{}_jp_{x+1} \cdot q_{x+j+1} + v^{n-1}{}_{n-1}p_{x+1} \right) \quad (\text{using } j = k - 1) \\ &= vq_x + vp_x (A_{x+1:\overline{n-1}|}^1 + {}_{n-1}E_{x+1}) \\ &= vq_x + vp_x A_{x+1:\overline{n-1}|} \end{aligned}$$

Recursive Relationships

There also exist recursive formulae for the second moments of the present value random variable, however, since the proofs are similar to those for the first moment, we only prove the case for a whole life insurance.

$$\boxed{{}^2A_x = v^2q_x + v^2p_x \cdot {}^2A_{x+1}} \quad \text{Whole life}$$

$$\begin{aligned} {}^2A_x &= \sum_{k=0}^{\infty} v^{2(k+1)} {}_kq_x = \sum_{k=0}^{\infty} v^{2k+2} {}_kp_x \cdot q_{x+k} \\ &= v^2q_x + \sum_{k=1}^{\infty} v^{2k+2} {}_kp_x \cdot q_{x+k} \\ &= v^2q_x + \sum_{j=0}^{\infty} v^{2(j+1)+2} {}_{j+1}p_x \cdot q_{x+j+1} \quad (\text{using } j = k - 1) \\ &= v^2q_x + \sum_{j=0}^{\infty} v^{2j+4} {}_jp_{x+1} \cdot p_x \cdot q_{x+j+1} \\ &= v^2q_x + v^2p_x \sum_{j=0}^{\infty} (v^{j+1})^2 {}_jp_{x+1} \cdot q_{x+j+1} \\ &= v^2q_x + v^2p_x \cdot {}^2A_{x+1} \end{aligned}$$

□

Recursive Relationships

For the other types of insurance the recursive formulae for the second moment are

$$\boxed{{}^2A_{x:\overline{n}|}^1 = v^2q_x + v^2p_x \cdot {}^2A_{x+1:\overline{n-1}|}^1} \quad \text{Term}$$

$$\boxed{{}^2A_{x:\overline{n}|} = v^2q_x + v^2p_x \cdot {}^2A_{x+1:\overline{n-1}|}} \quad \text{Endowment}$$

$$\boxed{{}^2_{n|}A_x = v^{2(n+1)} {}_np_x \cdot q_{x+n} + {}^2_{n+1|}A_x} \quad \text{Deferred}$$

Continuous Insurances

Policy	Notation for E(Z)	Formula for E(Z)	E(Z ²)
Whole life	$\bar{A}_x$	$\int_0^\infty v^t {}_t p_x \mu_{x+t} dt$	${}^2\bar{A}_x$
<i>n</i> -year term life	$\bar{A}_{x:\overline{n} }^1$	$\int_0^n v^t {}_t p_x \mu_{x+t} dt$	${}^2\bar{A}_{x:\overline{n} }^1$
<i>n</i> -year pure endowment	$A_{x:\overline{n} }^1$	$v^n {}_n p_x$	${}^2A_{x:\overline{n} }^1$
<i>n</i> -year endowment	$\bar{A}_{x:\overline{n} }$	$\bar{A}_{x:\overline{n} }^1 + A_{x:\overline{n} }^1$	${}^2\bar{A}_{x:\overline{n} }$
<i>n</i> -year deferred whole life	${}_n\bar{A}_x$	$\int_n^\infty v^t {}_t p_x \mu_{x+t} dt$	${}_n{}^2\bar{A}_x$

Discrete Insurances

Policy	Actuarial present value	
	Notation	Formula
Whole life	A_x	$\sum_{k=0}^{\infty} v^{k+1} {}_k p_x q_{x+k}$
<i>n</i> -year term life	$A_{x:\overline{n} }^1$	$\sum_{k=0}^{n-1} v^{k+1} {}_k p_x q_{x+k}$
<i>n</i> -year endowment	$A_{x:\overline{n} }$	$A_{x:\overline{n} }^1 + A_{x:\overline{n} }^1$
<i>n</i> -year deferred whole life	${}_n A_x$	$\sum_{k=n}^{\infty} v^{k+1} {}_k p_x q_{x+k}$

END